NOT GRADED, NOT OPTIONAL · WEEK 1

Your ungraded work

Week 1 · How physiology works and what keeps you steady

None of this carries points, and all of it is the route to the work that does. Print it, or work it on your own paper. Anything you cannot do from a blank page is the thing to take to your Mastery Check.

01 Retrieval target

Ten minutes, one sheet of paper, nothing open. Do this before anything else on the page.

Close everything. Reconstruct a complete negative feedback loop for body temperature on a cold day, from the stimulus all the way to the response, with every component named and the feedback arrow drawn.

02 Brain dumps

Three minutes each, blank page, everything you can get down. Then read the statement under it and mark, in a second color, what you left out. The second color is the useful part.

1 · PHYSIOLOGY AND LEVELS OF FUNCTION

→ Brain dump everything you have on **physiology and levels of function**. Three minutes.

Check against this: Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.

2 · STRUCTURE AND FUNCTION RELATIONSHIP LAB

→ Brain dump everything you have on **structure and function relationship**. Three minutes.

Check against this: Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.

3 · HOMEOSTASIS DEFINED

→ Brain dump everything you have on **homeostasis defined**. Three minutes.

Check against this: Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.

4 · FEEDBACK LOOP COMPONENTS LAB

→ Brain dump everything you have on **feedback loop components**. Three minutes.

Check against this: Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.

5 · NEGATIVE AND POSITIVE FEEDBACK

→ Brain dump everything you have on **negative and positive feedback**. Three minutes.

Check against this: Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.

6 · FEEDFORWARD CONTROL AND ACCLIMATIZATION

→ Brain dump everything you have on **feedforward control and acclimatization**. Three minutes.

Check against this: Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.

7 · LOCAL AND REFLEX CONTROL PATHWAYS

→ Brain dump everything you have on **local and reflex control pathways**. Three minutes.

Check against this: Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.

8 · MASS BALANCE LAB

→ Brain dump everything you have on **mass balance**. Three minutes.

Check against this: Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

9 · UNITS AND UNIT CONVERSION LAB

→ Brain dump everything you have on **units and unit conversion**. Three minutes.

Check against this: Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

10 · GRAPHING AND DATA INTERPRETATION LAB

→ Brain dump everything you have on **graphing and data interpretation**. Three minutes.

Check against this: Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

11 · EXPERIMENTAL DESIGN AND CONTROLS LAB

→ Brain dump everything you have on **experimental design and controls**. Three minutes.

Check against this: Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

12 · MEASUREMENT ERROR AND VARIABILITY LAB

→ Brain dump everything you have on **measurement error and variability**. Three minutes.

Check against this: Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

03 Draw it

Two prompts per competency. Do either one. Boxes, arrows and labels, not sentences, because a missing arrow shows up immediately and a missing sentence does not.

These are the same prompts as your printable note sheet, gathered here so you can see the whole week at once.

1 · PHYSIOLOGY AND LEVELS OF FUNCTION

A Build your own ladder. Pick a process, not oxygen delivery, and draw it at all six levels, molecule up to organism. One small box per level, labeled. Then break one rung and write, level by level, what stops working above it.

B Pick your own process, not oxygen delivery. Draw the same thing twice. First as anatomy sees it, the structures and their parts. Then as physiology sees it, the same thing working, with arrows for what moves and what changes. Under each drawing, one line naming the question it answers.

2 · STRUCTURE AND FUNCTION RELATIONSHIP LAB

A Draw one structure from this week and label the single feature that makes its job possible. Then sketch what would stop working if that feature changed.

B List three structures from this week. Beside each, write the one property its function depends on, in six words or fewer.

3 · HOMEOSTASIS DEFINED

A Write the definition of homeostasis, then underline the three words doing the real work. Beside each underlined word, one line on why it matters.

B Draw two graphs side by side, a value in steady state and a value at equilibrium. On each, label what is still moving and what it costs.

4 · FEEDBACK LOOP COMPONENTS LAB

A Draw the five component reflex pathway top down and label all five plus the feedback arrow. Labels only, no sentences.

B Fill in the five components for thermoregulation on a cold day. Then mark which component you would suspect first if the patient never shivered.

5 · NEGATIVE AND POSITIVE FEEDBACK

A Draw one negative feedback loop and one positive feedback loop as arrow diagrams. On each, mark what makes it stop.

B Draw clotting as a positive feedback loop, then show what ends it. Underneath, write one line on why the body can afford to use positive feedback here.

6 · FEEDFORWARD CONTROL AND ACCLIMATIZATION

A Draw a timeline of eating a meal. Mark where feedforward starts and where feedback takes over.

B Draw a setpoint as a horizontal line. Show what fever does to it, and mark at each stage what the patient feels.

7 · LOCAL AND REFLEX CONTROL PATHWAYS

A Draw a local control example and a reflex control example side by side. On each, mark how far the signal travels.

B Standing up suddenly: draw the reflex, label all five components, and write one line on why local control could not have handled it.

8 · MASS BALANCE LAB

A Write the mass balance equation and label every term. Then draw a body outline with arrows in and out for sodium, labeling which arrow is which term.

B Work a mass balance for one substance across one day. Then change a single term and show the new result underneath.

9 · UNITS AND UNIT CONVERSION LAB

A Take one real lab value and walk it through every unit physiology reports it in. Glucose in mg/dL to mmol/L, or a pressure in mmHg to kPa. Show the conversion as a drawn chain, one box per step, with the factor written on the arrow between them. Then write one line on which unit a clinician would actually say out loud, and why that one.

B Draw a number line for one quantity, say volume, and mark liters, milliliters and microlitres on it in the right places. Put three real physiological volumes on that line: a tidal breath, a drop of blood taken for a glucose stick, and the water in an adult body. Under it, write what goes wrong in a patient if you read one of them off by a factor of a thousand.

10 · GRAPHING AND DATA INTERPRETATION LAB

A Plot the mission patient's sodium across the three days. Label both axes with units and mark the reference range on the graph.

B Draw a saturation curve. Label the plateau, then write one line on what the plateau means physiologically.

11 · EXPERIMENTAL DESIGN AND CONTROLS LAB

A Sketch an experiment as three labeled boxes: what you changed, what you measured, and your control.

B Take a claim from this week and design the control group that would test it. Draw both groups and label the one difference between them.

12 · MEASUREMENT ERROR AND VARIABILITY LAB

A Draw two sets of repeated measurements, one with high variability and one with low. Mark which difference between readings is real.

B List three sources of variation in a single measurement. Beside each, one way to reduce it.

04 Book problems

Silverthorn, Chapter 1: questions 9, 13, 14, 16, 17, 18 and 19. The reasoning set: the body map, the feedback compare and contrast, why-does-blood-flow, and the four data and study design problems.

Find them at the end of the chapter in your eText, inside Access Pearson in Canvas.

Two routes, and both count.

1. Work it forwards if you have a way in. Get as far as you can before you look at anything.
2. If you open one and have no idea how to start, read the worked solution and write beside each line why that step is there. Then close it and reproduce the whole thing from blank paper. That counts every bit as much as the first route.

05 Teach it

Pick one competency from the list above and explain it out loud to an empty room, or record yourself, with nothing in front of you.

Where you stall is the gap. That is the whole diagnostic, and it is faster than any other one on this page.

If you get through it cleanly, do it again with the hardest competency on the list, not the easiest.

06 Then check yourself

Build a Mastery Check for this week. It is drawn fresh every time, so a second attempt measures what you learned rather than what you remembered, and it tells you which competencies to come back to.